METRIC

MIL-DTL-38999/23C
w/AMENDMENT 2
11 August 2022
SUPERSEDING
MIL-DTL-38999/23C
w/AMENDMENT 1
5 August 2016

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, RECEPTACLE, JAM-NUT MOUNTING, HERMETIC, HERMETIC SOLDER CONTACTS, SERIES III, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification sheet and MIL-DTL-38999.

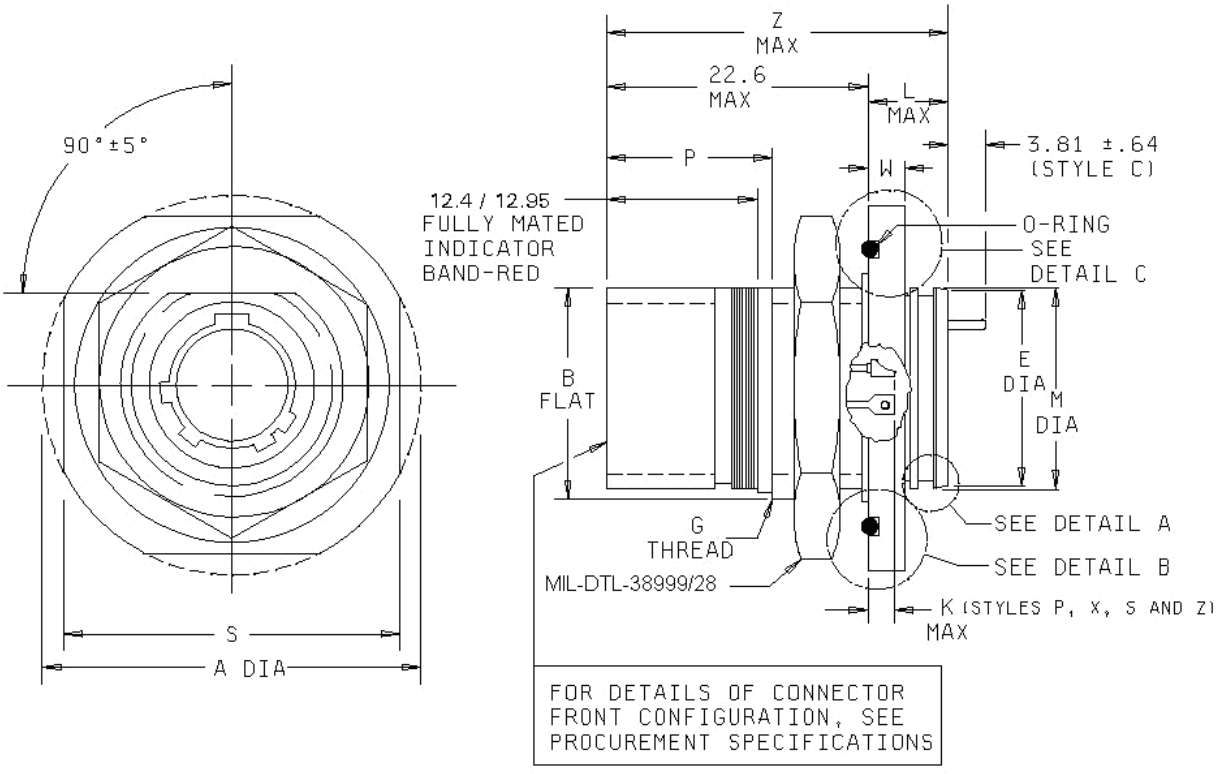
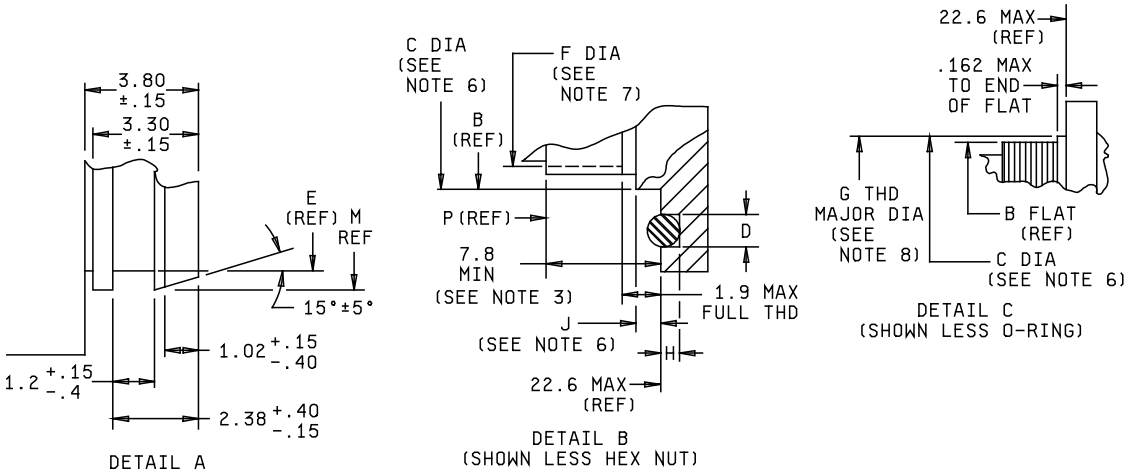


FIGURE 1. Receptacle, classes N and Y.

AMSC N/A
FSC 5935

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Shell size	Shell size code	A dia ± .3	B flat + .10 - .15	C dia + .1 - .2	D + .13 - .03	E dia ± .3	F dia max	G thread
9	A	30.2 (1.189)	16.53 (.651)	17.4 (.685)	2.39 (.094)	15.3 (.602)	15.9 (.626)	M17X1-6g 0.100R
11	B	34.9 (1.374)	19.07 (.751)	20.6 (.811)	2.39 (.094)	18.4 (.724)	18.8 (.740)	M20X1-6g 0.100R
13	C	38.1 (1.500)	23.82 (.938)	25.4 (1.000)	2.39 (.094)	21.6 (.850)	23.8 (.937)	M25X1-6g 0.100R
15	D	41.3 (1.626)	26.97 (1.062)	28.5 (1.122)	2.39 (.094)	24.8 (.976)	26.8 (1.055)	M28X1-6g 0.100R
17	E	44.5 (1.752)	30.15 (1.187)	See note 6	2.39 (.094)	28.0 (1.102)	30.8 (1.213)	M32X1-6g 0.100R (see note 2)
19	F	49.2 (1.937)	33.32 (1.312)	See note 6	3.58 (.141)	31.2 (1.228)	33.8 (1.331)	M35X1-6g 0.100R
21	G	52.4 (2.063)	36.50 (1.437)	See note 6	3.58 (.141)	34.3 (1.350)	36.8 (1.449)	M38X1-6g 0.100R
23	H	55.6 (2.189)	39.67 (1.562)	41.2 (1.622)	3.58 (.141)	37.5 (1.476)	39.8 (1.567)	M41X1-6g 0.100R
25	J	58.7 (2.311)	42.85 (1.687)	44.4 (1.748)	3.58 (.141)	40.7 (1.602)	42.8 (1.685)	M44X1-6g 0.100R

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FIGURE 1. Receptacle, classes N and Y- Continued.

Shell size	Shell size code	H ± .13	J + .0 - .71	K max		L max
				with pin	with socket	
9	A	1.14 (.045)	1.5 (.059)	5.3 (.209)	5.9 (.232)	9.1 (.358)
11	B	1.14 (.045)	1.5 (.059)	5.3 (.209)	5.9 (.232)	9.1 (.358)
13	C	1.14 (.045)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.1 (.358)
15	D	1.14 (.045)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.1 (.358)
17	E	1.14 (.045)	---	5.1 (.201)	5.7 (.224)	9.1 (.358)
19	F	1.91 (.075)	---	5.1 (.201)	5.7 (.224)	9.7 (.382)
21	G	1.91 (.075)	---	5.1 (.201)	5.7 (.224)	9.7 (.382)
23	H	1.91 (.075)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.7 (.382)
25	J	1.91 (.075)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.7 (.382)

FIGURE 1. Receptacle, classes N and Y - Continued.

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Shell size	Shell size code	M dia +.3, -.0	P +.4 -.1	S ±.4	W +.3 -.1	Z max	O-ring part number M25988/3 (see note 5)
9	A	16.3 (.642)	14.1 (.555)	27.0 (1.063)	2.6 (.102)	29.2 (1.150)	-019
11	B	19.4 (.764)	14.1 (.555)	31.8 (1.252)	2.6 (.102)	29.2 (1.150)	-022
13	C	22.7 (.894)	14.3 (.563)	34.9 (1.374)	2.6 (.102)	29.3 (1.154)	-024
15	D	25.9 (1.020)	14.3 (.563)	38.1 (1.500)	2.6 (.102)	29.3 (1.154)	-026
17	E	29.0 (1.142)	14.3 (.563)	41.3 (1.626)	2.6 (.102)	29.3 (1.154)	-028
19	F	32.2 (1.268)	14.3 (.563)	46.0 (1.811)	3.4 (.134)	30.1 (1.185)	-128
21	G	35.4 (1.394)	14.3 (.563)	49.2 (1.937)	3.4 (.134)	30.1 (1.185)	-130
23	H	38.6 (1.520)	14.3 (.563)	52.4 (2.063)	3.4 (.134)	30.1 (1.185)	-132
25	J	41.7 (1.642)	14.3 (.563)	55.6 (2.189)	3.4 (.134)	30.1 (1.185)	-134

NOTES:
1. Dimensions are in millimeters.
2. Modified major diameter: 31.95 - 31.80 mm.
3. Panel thickness: 3.2 - 1.58 mm.
4. Polarizing position N is shown.
5. O-rings: O-ring shall be in accordance with MIL-DTL-25988/3.
6. C and J dimensions define feature required for proper fit in type E panel.
7. F diameter defines thread start.
8. This dimension applies to shell sizes 17, 19, and 21 only (C diameter and G diameter are the same).
9. Inch equivalents are given for general information only.
10. The specified red band location is measured from the front edge of the shell to the back edge of the red band.

FIGURE 1. Receptacle, classes N and Y - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1. Interface dimensions shall conform to MIL-DTL-38999.

This connector mates with MIL-DTL-38999/26.

O-ring and hex nut shall be provided with jam-nut receptacles.

For insert arrangements: See MIL-STD-1560.

Connector accessories: See SAE-AS85049.

Part or Identifying Number (PIN) example:

D38999/	23	Y	A	35	P	N

D38999/	DoD number prefix
23	Specification sheet number
Y	Class
A	Shell size code
35	Insert arrangement
P	Contact style
N	Polarizing positions (N is required for normal position)

Eyelet contact termination: See MIL-DTL-38999.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-DTL-38999/28
MIL-DTL-38999/26
MIL-DTL-25988/3
MIL-STD-1560
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2022-181)

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force – 19

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